Verification matrix

English version of ../../verification/matrix.md.

Check	What it proves	Command	CI
Backend tests on SQLite	the logic needs no external database	`make test`	`ci.yml` → `backend`
ruff + mypy	static rules are respected	`make lint typecheck`	`ci.yml` → `backend`
Models ↔ migrations match	the schema does not drift from the code	`make alembic-check`	`ci.yml` → `backend`
Frontend build	the UI builds reproducibly	`make build-frontend`	`ci.yml` → `frontend`
Migrations on an empty database	installation from scratch is possible	`make alembic-upgrade`	`portability.yml` → `alembic-fresh`
Repeated `upgrade head`	the upgrade is idempotent	`make alembic-upgrade` twice	`portability.yml` → `alembic-upgrade-path`
Backup → restore → smoke	data is restored	`pg_dump` → `pg_restore` → `make smoke-test`	`portability.yml` → `backup-restore`
The full stack comes up	the services talk to each other	`make compose-up` + `make smoke-test`	`portability.yml` → `compose-smoke`
Start without integrations	GigaChat/Keycloak/S3 are optional	`make compose-up SERVICES="backend frontend"`	`portability.yml` → `config-without-secrets`
Documentation is consistent	links are alive, module specs are complete	`make docs-check`	`portability.yml` → `docs`
Host check	the environment is suitable before startup	`make doctor`	locally
Checks run on every branch	branch work does not reach merge unverified	`push` to any branch + PR	`ci.yml`, `portability.yml` (`branches: ["**"]`)
The CI mirror has not drifted from GitHub	the result does not depend on the platform	comparing the steps of `.sourcecraft/ci.yaml` with `ci.yml`	manually on CI changes

What is not checked automatically

- Real LMS/CMS exchange: the adapter is a mock, the contract is covered by dry-run tests.
- Performance and load.
- The Yandex Cloud production contour: verified by hand following operations/deployment.md.
- Parity of the two CI descriptions (`.github/workflows/ci.yml` and `.sourcecraft/ci.yaml`) by step set — a divergence is possible, see the audit.